AUTHOR TEMPLATE · MAXIMUM 300 WORDS

Structured abstract example

Illustrative example only. Replace all example names, study details, ethics approvals and findings with your own research.

Molecular Dissection of Neuroinflammatory Signaling in Ischemic Stroke

Amit K. Sharma[1]*, Neha Rastogi[1], Rajeshwar S. Verma[2]

[1] Department of Pharmacology & Toxicology, NIPER-Raebareli, Lucknow 226002, India.
[2] Division of Neuroscience & Ageing Biology, CSIR-CDRI, Lucknow 226031, India.

* Corresponding author: [your email] | Presenting author: [underline the name]

Track: TR-01 — Neuropharmacology & Neurotherapeutics

Background & objectives

Ischemic cerebral stroke initiates acute microglial polarization accompanied by catastrophic neurovascular breakdown. Despite recanalization therapies, post-ischemic secondary neuroinflammation remains therapeutically unaddressed. Here, we hypothesized that pharmacological inhibition of NLRP3 inflammasome priming via a novel small-molecule scaffold suppresses neurotoxic cytokines and salvages ischemic penumbral tissue.

Materials & methods

Adult male Sprague-Dawley rats (250–280 g, IAEC Approval # NIPER/IAEC/2026/04) were subjected to 90-minute transient middle cerebral artery occlusion (tMCAO) followed by 24h reperfusion. Vehicle or Test Agent (5, 10, 20 mg/kg i.v.) was administered at reperfusion onset. Neurological deficit scoring (mNSS), TTC cerebral infarct planimetry, blood-brain barrier permeability (Evans blue extravasation), and immunofluorescence for Iba-1/GFAP were quantified.

Results & discussion

Test Agent (10 mg/kg) caused a 48.6% reduction in total cerebral infarct volume compared to vehicle-treated controls ($p < 0.001$, one-way ANOVA with Dunnett's post-test). Neurological deficit scores improved significantly within 24 hours (mNSS 5.2 vs 11.8 in vehicle). Western blotting of penumbral tissue confirmed downregulation of cleaved caspase-1 and mature IL-1beta without altering systemic hemodynamics.

Conclusions

The findings establish this novel pharmacophore as a potent, brain-penetrant neurovascular protective lead candidate capable of dampening neuroinflammation during the hyperacute reperfusion window.

Keywords: Ischemic stroke; NLRP3 inflammasome; Microglia; Neuroprotection; NIPER-Raebareli

Presentation preference: [ ] Oral presentation [ ] Poster presentation